Applied Soft Computing 71 (2018) 66–77

Contents lists available at ScienceDirect

Applied Soft Computing

journal homepage: www.elsevier.com/locate/asoc

Securing the operations in SCADA-IoT platform based industrial control system using ensemble of deep belief networks

Shamsul Huda[a,*], John Yearwood[a], Mohammad Mehedi Hassan[b], Ahmad Almogren[b]

[a] Deakin University, School of IT, Geelong, Australia
[b] College of Computer and Information Sciences, King Saud University, Riyadh, 11543, Saudi Arabia

ARTICLE INFO

Article history:
Received 6 January 2018
Received in revised form 9 May 2018
Accepted 4 June 2018
Available online 26 June 2018

Keywords:
IoT
SCADA network
Industrial control systems
Malicious attack
Deep belief network
Man-in-the-middle attack
Critical infrastructure

ABSTRACT

Internet of Things (IoTs) platform is increasingly being used in modern industries. Billions of devices with smart sensing capabilities, PLCs, actuators, intelligent electronic devices (IEDs) of industrial control systems (ICS) and supervisory control and data acquisition (SCADA) network are connected over IoT platform. IoT platform has facilitated modern industries an efficient monitoring and controlling of physical systems (various hardware and machineries) resulting in an intelligent data acquisition, processing and highly productive and profitable management of business. Initially, these devices have been deployed without any security concern considering these will run in isolated networks. With the new IoT platform scenario, SCADA based ICS networks are integrated with the corporate networks over the internet. Therefore, the devices of a SCADA network are facing significant threat of malicious attacks either through the vulnerabilities of the corporate network or the devices used in the SCADA. Traditional IT security software products are not enough for ICS as these software products consider only operating system related calls and application program interface (API) behaviour of applications, which are only focused on corporate business solutions and related technologies. In this paper, we propose a secure architecture for ICS network that proposes a detection model based on SCADA network traffic. The proposed architecture develops two ensembles based detection algorithms using deep belief network (DBN) and standard classifier, including support vector machines (SVM). The novelty of the proposed architecture is that it uses network traffic feature and payload feature for detection model instead of conventional signature based or API based malware detection technique. In addition, ensemble-DBN of the proposed architecture can overcome many limitations of standard techniques, including the complexity and big size of the training data.

The proposed architecture for ICS has been verified using a real SCADA network data. Experimental results show that our ensemble based detection system outperforms over existing attack detection engines.

© 2018 Elsevier B.V. All rights reserved.

1. Introduction

In today's smart industry arena, Internet of Things (IoTs) are extensively integrated with industrial corporate network to increase the computation and storage capabilities, to monitor and control the physical systems from remote locations for high productivity of the industries and maximize the economic benefits [1–4]. Industrial IoTs can boost revenues significantly by maximizing the productivity through the intelligent business strategies and transformation of workforces, cheap data communication and intelligent data processing techniques and integration of those with the industrial control systems (ICS).

IoT devices enable collection, aggregation of data and make that available to the corporate business for intelligent analysis nd maximizing the throughput of entire operations resulting in a significant boost to the revenue of the companies.

As mentioned in Fig. 1, in modern industrial systems (Supervisory control and data acquisition, SCADA), the control networks are composed of programmable logic controller (PLCs), Remote Terminal Units (RTUs), intelligent electronic devices (IEDs). These field devices collect information from the smart sensors attached with the physical systems. Based on that information and locally installed control logic, local controllers (PLC/ RTU) directthe actuators in the IEDs to perform certain operations on the physical

* Corresponding author.
E-mail addresses: shamsul.huda@deakin.edu.au (S. Huda), j.yearwood@deakin.edu.au (J. Yearwood), mmhassan@ksu.edu.sa (M.M. Hassan), ahalmogren@ksu.edu.sa (A. Almogren).

https://doi.org/10.1016/j.asoc.2018.06.017
1568-4946/© 2018 Elsevier B.V. All rights reserved.

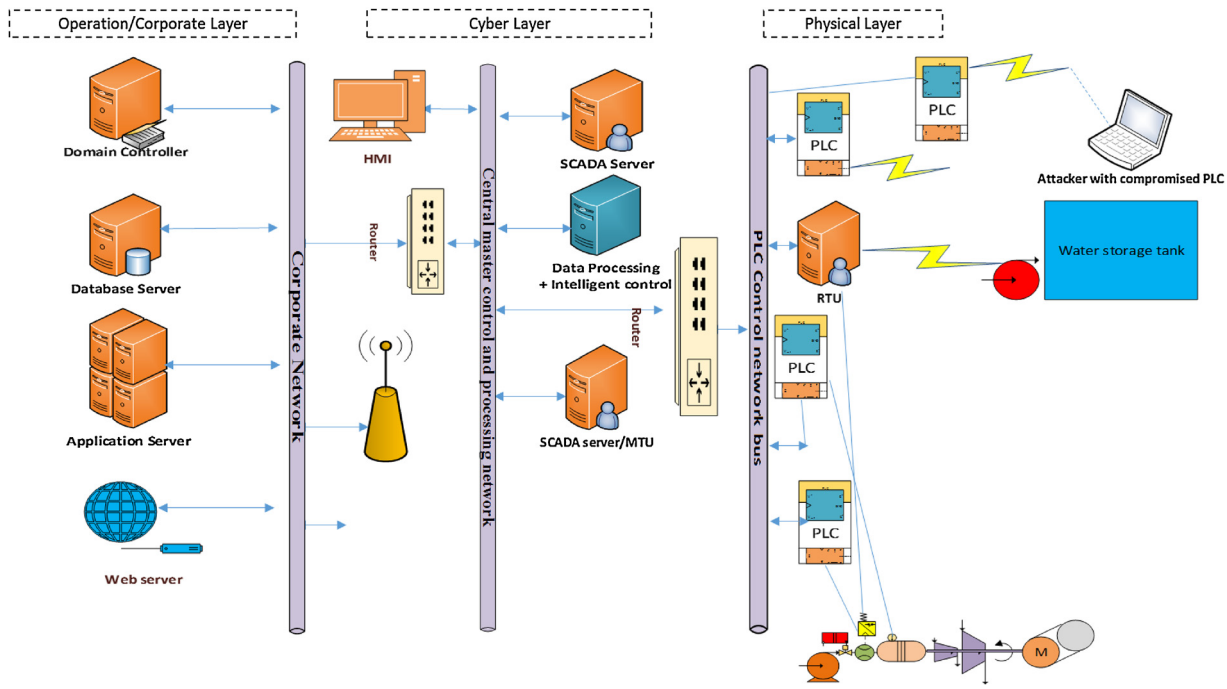

Fig. 1. A model of SCADA based industrial control network integrated with the corporate network.

systems. The servers in the process layer monitor and control the remote devices, supervise the field devices and provide distributed decisions on a safety critical situation [2,5].

Due to tremendous growth of lessexpensive IoT technologies and platforms, conventional isolated SCADA networks are extended and integrated with the corporate business networks and IT systems. Numerous devices with smart sensing capabilities are connected to SCADA based industrial control system (ICS) over the IoT platform. This facilitates a less-expensive data acquisition from machines/physical system and production levels, execution of business analytics on the collected low level data, faster automated mining of the data and integration of intelligent decisions to the control network for efficient management of the production systems. However, this integration exposes the field devices of SCADA network to severe security threats [11–14].

Benefits of traditional IT security systems and related anti-malware tools and their limitations for cyber-attacks on IoT-SCADA networks were reviewed in [7–10]. Current generation anti-malware tools are designed based on the behaviour of executable files, which are focused on the operating system (OS) calls and Application program interfaces (APIs) [15]. These are more suitable to safeguard applications and systems, which operate under the corporate network [15] and use extensively the OS calls and APIs. In contrast, as mentioned in Fig. 1, SCADA based industrial networks operate on many different protocols including MODBUS, Distributed Network Protocol version 3 (DNP3), EtherNet/IP and are interconnected with PLCs, RTU, master terminal unit (MTU), human machine interfaces (HMI), sensors and actuators. RTUs are used to monitor the states of a physical device locally, to store control parameters and run a control program to control the states of physical devices. Master Terminal Units (MTU) are located centrally to observe and control different RTUs. MTUs are connected with RTUs via the communication link. MTUs communicate the RTUs to read the state parameters of physical devices located in remote places and read the control parameters at RTUs. Human Machine Interfaces (HMIs) are used to display these parameters at MTUs where human operators can visualize the states of physical devices using the parameters. HMIs enable operators to change the parameters for RTUs. The data communication between MTUs, RTUs, HMI and physical devices happens mostly on physical, data and application layers. The physical layer can be wireless or wired. Obviously, network traffic and application behaviour in industrial control systems are different from the applications used in traditional IT systems.

Existing corporate network drags more security holes for industrial network through its own vulnerabilities related to access password, which later can be used by the attackers to access the control network. The IEDs in substations exports/imports the configuration file of substation, which is known as SCL (substation configuration language) providing the device information. Attackers can get the password through different cracking techniques. Then attackers can gain access to the critical data of distribution networks such as topologies of the network, device information and status of the actuators. This can provide enough information to initiate attacks on multiple substations simultaneously, which can create a total black out [36] in the distribution network.

Due to lack of awareness and security expertise in control software and operating systems (OS), limited opportunity of upgrade-ability, incompatibility of control-software with new versions of OS [11–17], IoT based industrial systems are forced to be operated on many older versions of OS or no-longer-supported version of OS and security patches and services packs [11–13]. The protocols used in the communications of industrial IoT based networks also have less security feature enabled due to the fact that IoT devices have less computational power. Therefore, the protocols are more vulnerable to malicious attacks. Once MTUs or RTUs are infected by the malicious software, due to the different types of network traffics and application behaviours from conventional IT system applications, traditional IT security tools (signature and API based anti-malware tools [15,18,19]) may fail to detect the malicious attacks. Therefore, to develop a secure IoT architecture and malicious attack detection system for industrial control system is well motivated and crucial for modern industries.

In this paper, we propose a security architecture and an attack detection model for IoT based industrial systems that can identify malicious attacks based on the network traffic and content of the traffic instead of API behaviour or signature based techniques

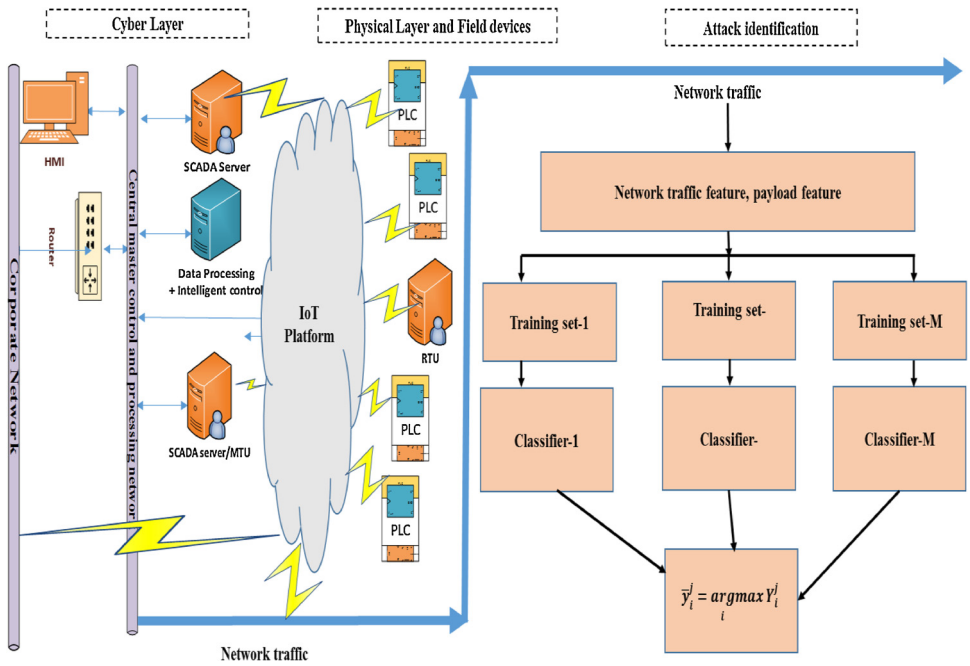

Fig. 2. Proposed ensemble based secure architecture for SCADA-IoT based industrial control system.

used in conventional anti-malware tools [15,18,19]. The proposed attack detection model uses different types of content features including readings from sensors of physical systems and values of control parameters; and network traffic. Network traffic includes the addresses of physical devices, packet length, standard code of the functions used in the control systems.

We propose an ensemble framework for the detection system. The proposed ensemble framework consists of two ensemble learning techniques based on the conventional machine learning classifier and deep learning.

Deep learning has become much popular due to its powerful capability to learn the hidden feature-patterns from the data. Deep belief network (DBN) [20–22] has been used in many complex pattern recognition problems including emotion recognition, image and video processing and classification [43–45].

However, it has been addressed in many articles [21,22] that the DBN shows its superior performance if the configuration of DBN is set appropriately. Configuration of a DBN requires appropriate settings for the hyper-parameters and DBN structures. The hyper-parameters of a DBN are the size of mini batch, initial settings of weight, number of epochs, learning rate, momentum and number of hidden layers and units. Settings of the hyper parameters could be crucial for the cases when insufficient data is used for training. Therefore, performance of a DBN may vary and can be degraded depending on the configurations. We propose different DBN structures with varying configurations and then use the ensemble of DBNs to develop a detection system.

The novelty and contributions of the paper are described in the following:

1 To develop a malicious attack detection model based on network traffic feature and payload feature for SCADA based industrial control system.
2 To develop a high performance attack detection system using advanced ensemble learning technique based on the deep learning and standard machine learning classifiers.
3 To justify the performance of the proposed approaches using a real control system environment.

The organization of the paper including the main topics of different sections are as follows. The second section of the paper is the related work which describes the current status of the problem in the literature and different attacks to an industrial control system. The third section describes proposed methodology including the ensemble learning based on deep learning and support vector machine classifier. The fourth section describes the data and experimental results. The last section describes the conclusions of this research and future directions.

2. Related work

Rapid technological advancement in the modern industries facilitates aggregation and transmission of the factory data to the corporate world through an integration of IoT and industrial control networks (ICSs). This provides intelligent business management and highly productive real-time control of the plant. However, this results into a major security threat for SCADA networks. Several authors [11–14,25–29] have reported the security challenges of integration of IoT platform and SCADA networks or integration of SCADA networks with the corporate networks. Johnson, C [25] explains the difference between isolated SCADA network and IoT-SCADA integration to the corporate network, risk of filed devices and air-gap problem for ICSs. Several examples have been provided for this type of integration including Tianjin, China smart city project [25]. In [26] authors conducted an extensive study to identify the security threats to the ICS network and non-functional requirements of the detection system for ICS. In [27], authors have explored the relationships between mechatronics, cyber physical system and IoT and their transitions. Authors in [27] proposed a resilient architecture which virtualizes the sensor networks and needs to be separated from the filed devices networks as sensor networks communicating over the internet. Zach DeSmita [28] investigated the cyber risk and vulnerabilities assessment methods for manufacturing industries and proposed a decision tree based intersection mapping technique to identify and categorize the security risks. In [47], Wei Gao also investigated a decision tree based model for attack detection in SCADA network. In [30], C.W. Johnson and M. Saleem have explored different types of firmware

attack on SCADA network. Their investigation shows that vendors make firmware code available over internet for customers which are downloaded and then penetration tested to change the code by the attackers. When corporate networks are compromised and attackers gain access to the ICSs, these changed code can be installed to the filed devices. There are several other potential risks installing firmware form internet sources such as man-in-the-middle attack which can install malformed firmware to the ICS. C.W. Johnson and M. Saleem [30] proposed hashing tools for firmware verification and authentication if the code is changed from vendor sources. Cristina Alcaraz et al. [31] investigated cyber potential attacks on CPS enabled power stations. They proposed different integration strategies of sensor network of ICS with the Internet including gateway solution, topology based integration or hybrid strategies to protect ICS network from malicious attacks. Ángel Manuel et al. [32] proposed supervised classification technique to identify cyber-attack on critical infrastructure such industrial robots. Abdulmohsen Almalawi et al. [33] proposed unsupervised method to develop a proximity based detection rule from different states of SCADA network. Igor Nai Fovino et al. [34] investigated ICT malware attack on SCADA system using 'MALsim' toolkit. Many researches were proposed to identify the vulnerabilities of IoT-SCADA platforms. Many architectures have been proposed to protect ICS networks from the compromised corporate ICT networks. Different authors also proposed supervised or unsupervised detection methods as mentioned above.

Usually, the central energy management system (EMS-which controls the whole power system and monitors), and remote RTUs communicate using DNP3. Communication can be made by wireless, radio frequency or dial-up transmission medium. Communication between SCADA station and substation can happen to use DNP3. DNP3 provides a method to identify remote devices' parameters and to retrieve new information [37–39]. DNP3 also provides some degree of security by authentication using unique session key, however, man-in-the-middle (MitM) attack can happen to extract port addresses. These can be used for denial of service (DoS) attack [26–29], manipulate time synchronization data to disrupt the synchronization and communication, stopping the acknowledgement messages to force the system in a continuous retransmission.

Modbus-TCP protocol is used between master PLCs/RTUs and slave devices or between PLCs and HMIs. However, there are a number of security concerns existing in this protocol. This protocol requires only Modbus addresses, function code and data. There is no verification that source or destination addresses are legitimated or not. Commands and data are sent using plain text and can be easily captured and spoofed since no encryption. DoS attacks can be easily implemented since there is no legitimate address checking [29–33,38].

Inter-Control Center Communication Protocol (ICCP) [26–33] is used to communicate between control centers such as EMS-SCADA or Regional SCADA centers. The protocol is used to access information including email message, energy market information, exception conditions, configuration of remote devices, and control of remote devices. There are a number of security concerns existing in ICCP including lack of authentication and encryption. Therefore, session hijacking, spoofing attack can be implemented on ICCP communications. ICCP works on wide areas and susceptible to DoS due to the exposure to the public network [30–34,37].

In summary, industrial protocols used in SCADA do not include basic security features such as authentication and encryption. All of these protocols are susceptible to security attacks when exposed to the corporate networks [40,41]. Recent trend in architecture integration [46] of IoT, cloud computing and SCADA network also makes easier for attackers to attack SCADA network. Authors in the literature [28–47], mostly investigated laboratory-based small scale SCADA system for attack detection and used conventional machine learning techniques such as decision tree, Bayesian classification or support vector machine based classification techniques [28–32]. Since the integration of IoT and SCADA connects billions of devices generating extremely large volume of SCADA traffic over the internet. Literature has less coverage on the impact of integration and related remedies, complexities of analysing a large volume of network traffic and detection approaches. Therefore, it is important to investigate how SCADA networks can be protected from the analysis of generated traffic, which can cover a very large heterogeneous business-industrial network. In this work, we focus on this gap and investigate techniques that can handle a large volume of SCADA network traffic to detect sophisticated attack patterns.

3. Proposed methodology

The proposed IoT based industrial control system introduces a secure integration model for SCADA connecting with the corporate network. This has been accomplished by developing an ensemble based detection model. Two main types of the detection model are proposed.

1 Ensemble of classifiers
2 Ensemble of Deep Belief Network (DBN)

3.1. Proposed ensemble of classifiers

Fig. 2 presents the proposed secure architecture which is able to monitor the network traffic from the cyber layer to physical layer or vice-versa. Any query from MTU to RTU/PLC or response from PLC/RTU to MTU is forwarded to the detection model. The traffic has two main components: traffic related component and payload related components. Traffic related components are the addresses of different field devices including PLC/RTU/MTU, Intelligent electronic devices (IEDs), the code for the function that is used for control purposes, length of each packet, packet error related data and interval between two packets. Payload related components are the data received from different measurements of sensors of physical devices, supervision-control/command/decision-inputs and operational mode related data.

The data from network traffic are used in an ensemble of classifiers as mentioned in Fig. 2. An ensemble learning builds a set of classifiers. Each classifier in the developed set is trained using a different data set. The final decision is made by taking the average value of the decision or by using a voting technique. The data set for each classifier can be made by using different techniques including a bagging or a boosting technique.

Let us consider a set of captured network traffic samples $\mathbf{V}$ which has both components of features as mentioned above. $v^j = \{v_1, v_2, v_3, \ldots\ldots v_m\}$ is a sample traffic instance and $v^j \in \mathbf{V}$. Each sample instance in $\mathbf{V}$ is identified by a class label is known as $y^j \in \{y_1, y_2, \ldots y_l\}$.

Let us assume $|\mathbf{V}| = \mathbf{N}$, bagging techniques divides the data set $\mathbf{V}$ by taking 'M' bootstrap sampled set $\{\mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3, \mathbf{V}_4, \ldots \mathbf{V}_M\}$. For each sampled set $\mathbf{V}_i$ is sampled randomly from $\mathbf{V}$ with replacement where $|\mathbf{V}_i| = L$

Each set $\mathbf{V}_i$ is used to train an independent classifier which learns the mapping function $F_{\mathbf{i}}$ that maps the network traffic feature space to the attack type space. Each new traffic sample n^j can be categorized using the function$F_{\mathbf{i}}$ as blow:

$$y_i^j \in \{y_1, y_2, \ldots y_l\} = F_{\mathbf{i}}\left(\mathbf{V}_i,\ v^j\right) \tag{1}$$

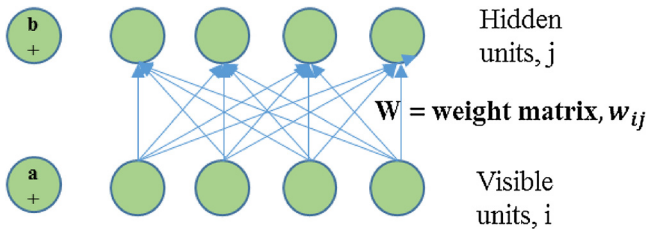

Fig. 3. Stochastic network representing RBM layer.

The final classification of sample n^j is decided based on a majority voting scheme as below

$$\bar{y}_i^j = \underset{i}{\mathbf{argmax}} Y_i^j \tag{2}$$

In this research, we used support vector machine (SVM) [36] as the base classifier in the ensemble learning model.

3.2. Proposed ensemble of deep belief network (DBN)

Deep learning [20] is a popular approach to extract the intrinsic hidden structure of data from a particular domain. The deep learning technique learns the intrinsic hidden structure of the data through multiple layers of abstraction where each layer transforms the raw input data to a slightly higher abstraction of input data by using some non-linear transformation. Geoffrey Hinton [21,22] has proven that many compositions of such non-linear transformations (abstraction of input data) can learn complex functions that easily can map a very large input space to the target label space. In addition, transformation of higher layer provides more discrimination abilities for detection. Deep belief network (DBN) [20–22] is one of the deep learning techniques. Core of DBN is the Restricted Boltzman Machine (RBM) [21] which is a generative model. The training of DBN is based on RBM training, which is trained using contrastive divergence procedure [22].

RBMs are used to represent the training samples, in our case the sample of network traffic data. Each RBM layer can be modelled using a two-layer network as presented in Fig. 3. Here the traffic features are the visible units denoted as 'i-th layer' and the generative model, which generates the features are the hidden states 'j'. Hidden units 'j' are also called the feature detector.

For the generative model in RBM, joint configuration of visible and hidden units (v, h) holds an energy [23] as below:

$$E(v, h) = -\sum_i a_i v_i - \sum_j b_j h_j - \sum_{ij} v_i h_j w_{ij} \tag{3}$$

a_i and b_j are the biases of visible and hidden units.

Then every possible pair of a visible and a hidden layer in the RBM network through the distribution

$$p(v, h) = \frac{1}{X} e^{-E(v,h)} \tag{4}$$

Here $X = \sum_{jj} e^{-E(v,h)}$. Each node in RBM layers can generate a random values of either 1 or 0 following the layer-wise conditional distribution which can be achieved from marginal distribution of Eq. (4), as mentioned below:

$$P\ v_{i=1} | h) = \sigma(a_i + \sum_j w_{ij} h_j) \tag{5}$$

$$P\ h_{j=1} \big| b) = \sigma(b_j + \sum_i w_{ij} v_i) \tag{6}$$

Where $\sigma(x) = \frac{1}{(1+e^{-x})}$, for Eq. (5), $x = a_i + \sum_j w_{ij} h_j$ and for Eq. (6), $x = b_j + \sum_i w_{ij} v_i$

The training procedure follows a stochastic steepest ascent algorithm which can follow a network weight update rule derived from the derivative of the log probability of marginal distribution $p(v) = \frac{1}{X} \sum_j e^{-E(v,h)}$.

The network weight are updated [20–23] as following Eq. (7)

$$\Delta w_{ij} = \varepsilon(< v_i h_j >_{data} - < v_i h_j >_{reconstruction}) \tag{7}$$

$< v_i h_j >_{data}$ is the expectation of the distribution of the data, $< v_i h_j >_{reconstruction}$) is the expectation of the distribution of reconstruction, $\varepsilon > 0$ is the learning rate and Δw_{ij} is the Change of weight matrix.

A DBN is constructed with the composition of many RBM layers as mentioned in Fig. 4. Once the RBM's are trained, the hidden weights 'W' are used to initialize a neural network (NN) of similar hidden structure of DBN. An output layer is added to the NN which has nodes depending on the number of class values. For a binary classification problem, the number of output nodes will be two. For our malicious attack detection problem, there are total seven attack types plus one normal traffic. Therefore, we choose an output layer of eight nodes. The NN is trained with a back propagation training algorithm.

The appropriate number of hidden layers which are required for a DBN is one of the most crucial research questions. The answer for this question depends on the problem domain and the complexity of the dataset. Usually for big training data sets with highly redundant samples, fewer parameters are required. Therefore, total hidden units and layers will be less. This rough estimation could impact the performance of resulting NN. Therefore, we propose an ensemble of DBNs.

In Fig. 4, the proposed architecture of the ensemble DBN is presented. We used three different structures of DBN. The first DBN in Fig. 4 has the equal number of nodes in each hidden layer. This has a standard number of hidden units, which are globally connected. It has several orders of magnitude higher than the required number of bits to specify the different types of attack. Then second type of DBN has less hidden units in the last layer than the first hidden layer. The third type of DBN is the opposite to the second DBN. Each DBN is trained separately and then is converted to a similar structure NN with the output layer added for an appropriate number of nodes. The second and third type of DBNs have fewer numbers of hidden units compared to the first DBN. These types of DBNs are considered to capture the inherent pattern of a big training set with highly redundant patterns. These DBNs have fewer parameters. The second DBN does a feature compression through the selection of more important features. While, the third DBN combines both feature compression and then representing the features using a higher number of parameters at the last hidden layer. A cross validation is used to train DBN plus NN, then testing is performed as part of the cross validation. The final decision is constructed by using a majority voting scheme following Eq. (2).

4. Experiments, results and discussion

Data set: The proposed architecture is verified with a real SCADA network data set provided by author Wei Gao [24,47]. The data set has samples of SCADA traffic from industrial control systems. Different attack types are described below. Detail description of data can be found in the article [24]. There are seven different types of attacks in the traffic samples of the data set.

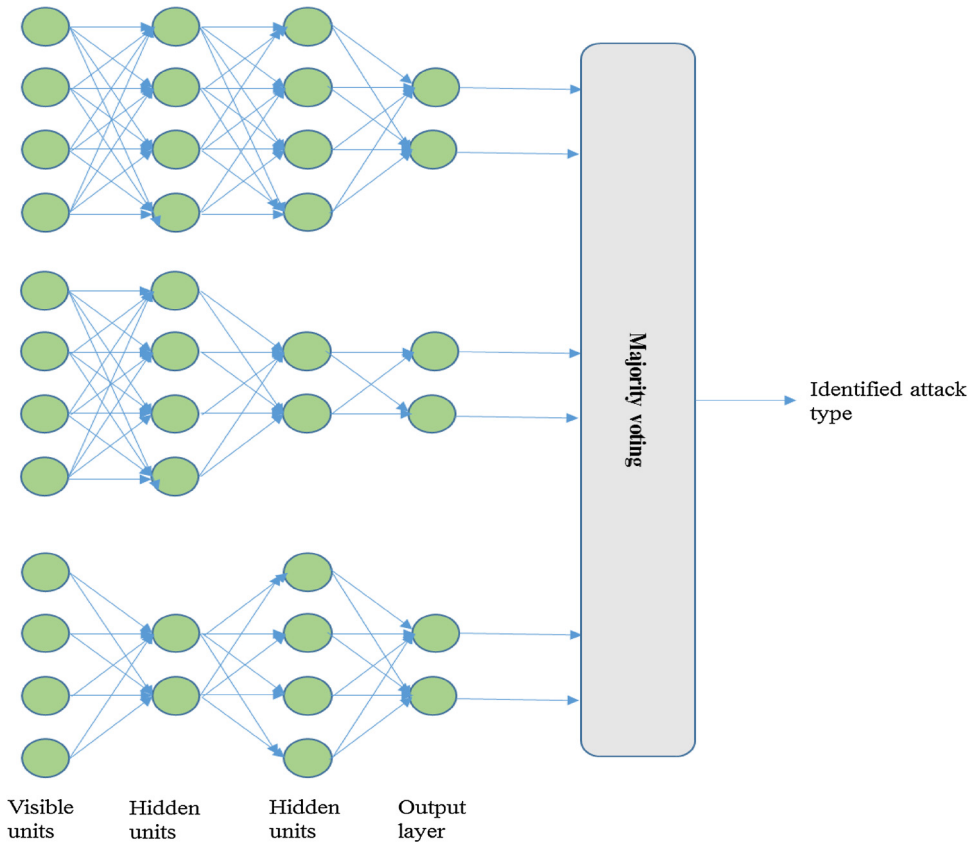

Fig. 4. Proposed ensemble-DBN based secure architecture for SCADA-IoT based industrial control system.

1) Reconnaissance Attack: Using this type of attack, an attacker tries to gather information about the network architecture such as supported network protocols, addresses of the connected servers, supported network operations to the SCADA servers, characteristics of the connected field devices including name of the device manufacturers, model number, memory map of the devices and system memory map.
2) Once an attacker gathers network information and devices information, he/she can use these information for more sophisticated attacks. Some of those attacks are described below:
   a Malicious Response Injection Attack (MRIA): Attacker can achieve username and password from infected corporate network which is used to access the SCADA network where the response packets transmitted from RTU to MTU during polling are modified and forwarded to the MTU. Often a denial of service (DoS) attack is performed to target RTU to delay its response and a third part device response is sent to the MTU during polling.
   b Naive Malicious Response Injection (NMRI): Here an attacker may have information about the network's servers and devices, but are not aware of detail of operations performed on the devices. So an attacker injects random invalid information to the packet.
   c Complex Malicious Response Injection (CMRI) attacks: In this type of attack, an attacker has full knowledge and understanding of SCADA network and devices including the operations, and states of the system, sensor measurements. An attacker modifies the packet such that it hides the real states of devices and the physical systems. This can be done in many ways including sending the same sensor reading to the MTU to mislead the operator about a real situation of a physical system. The other types can be a simulation of the system for a fault condition and estimate the measurements for faulty situation. Then modify the readings in the packet with estimated value to mislead the operator that a fault happened. This requires complete knowledge of the system which can be achieved through reconnaissance attack.
   d Malicious Command Injection Attack: In this category, an attacker may inject unauthorized commands and control information to the supervisory command packet. Often human operator sends supervisory command and control information to the RTU, PLC and IEDs to reset some control settings or logic at the target filed devices. Normally, target devices execute locally installed logic and setting to monitor and control the physical systems. But supervisory command can override these local logic and settings. An attacker can abuse this protocol between MTU and RTU by injecting wrong command in the packet. This can impact the system starting from simple interruption of operation to destruction of a whole system. This can be three types as: (Malicious State Command Injection (MSCI), Malicious Parameter Command Injection (MPCI), and Malicious Function Code Injection (MFCI)).
   e Malicious State Command Injection (MSCI): Actuators operate on a physical system directly to control the operations of the physical system such as ON/OFF. The actuators get the command from RTU. Usually actuators read the command from register of RTU, which is pre-set or received from MTU. An attacker can send MSCI attack to change the settings in the RTU register, which will mislead the actuators to operate the physical system incorrectly.
   f Malicious Parameter Command Injection (MPCI): Filed devices in SCADA network such as PLCs operate based on installed ladder logic and set points parameters (parameters

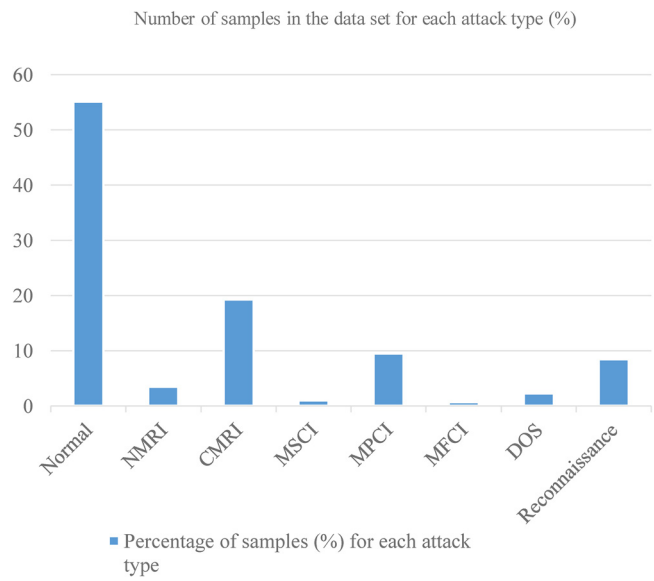

Fig. 5. Distribution of attack samples and normal traffic in the data set.

for PID controller). This type of attack changes the parameters stored into PLC registers to incorrect values so that controller malfunctions or does not function at all.

g Malicious Function Code Injection (MFCI): Many application layer function codes have been provided by the manufacturers for diagnostic purposes. If an attacker has knowledge of these, he/she can abuse of such function codes. An example is a MODBUS function 'force' which forces MODBUS server in listen only mode and no response against the polling by a human operator. MFCI attack will place the MODBUS server out of visibility and control.

h Denial of Service Attacks (DoSs): This attack is performed to stop part of the SACDA network or any particular RTU/ field devices. This can be done by sending a large number of packets to the target device faster than its processing time. Sometime an attacker may modify a packet and send to the filed device such that it causes run-time error resulting in a crash of the program or operating system.

The data set provided for gas pipeline system (ICS) [24,47] has a total of 97,018 samples of instances from SCADA network traffic. Within this set, 61,155 instances are normal traffic, and the rest are attacks. Since this has a high number of normal samples, which make the class distribution imbalanced. In our experiments, we discarded some normal samples. First, 17,018 samples have been discarded and total of 80,000 samples are considered for analysis. Fig. 5 presents the statistics of different attack types in the data set.

The performance of proposed detection approaches have been justified based on four different metrics including accuracies, True positive rate (TPR), false positive rate (FPR), and Precision.

The base classifier for first approach 'ensemble of classifier' is SVM. Three different types of DBN are used for second approach 'Ensemble of DBN'. Three different types are:

1 DBN (equal hidden units in all layers), here we use 100 hidden units for first and second layer of RBMs.
2 DBN (last layer less hidden units), here we use 100 hidden units for first layer and 50 for last layer of RBM.
3 DBN (First layer less hidden units), here we use 50 for first layer and 100 for last layer of RBM.

Table 1
Accuracies of cross-validation for different detection approaches.

Detection approaches	Accuracies (%)
SVM	93.88
Ensemble of SVMs	94.41
DBN (equal hidden units in all layers)	94.65
DBN (last layer less hidden units)	94.62
DBN (First layer less hidden units)	94.60
Ensemble of DBNs	95.60
Decision Tree based approach [24,47]	93.1

DBN training has the epoch of 1000 and NN training has 2000 epochs. All DBN has been tested separately on a 10 fold cross validation. SVM and Ensemble of SVM have also been tested using a 10-fold cross validation. Experimental results are described in (Figs. 6–11) and Tables 1–4.

Figs. 6–8 present the reconstruction error rate of RBM layers for different types of DBN used in our proposed methods. From reconstruction error graphs, it is seen that, when using an equal number of hidden units reconstruction error for both RBM layers archive the lowest reconstruction error rate (Fig. 8) compared to other two structures of DBN as mentioned in Figs. 6 and 7.

ANN training error (MSE) for each structure (DBN (First layer less hidden units), DBN (last layer less hidden units) and DBN (equal hidden units in all layers)) are presented in Figs. 9–11. The DBN structure 'DBN (last layer less hidden units)' in Fig. 10 achieves better error rate than 'DBN (First layer less hidden units)' in Fig. 9. However, DBN (equal hidden units in all layers) achieves lowest MSE in Fig. 11.

Table 1 shows that proposed ensemble of SVM achieved higher accuracies than standard classifier SVM without ensemble. Three different structure of DBNs achieved accuracies (94.65%, 94.62% and 94.60%). However ensemble of DBNs achieved the highest accuracies. This experiment shows that DBN structure can affect the performance of classifier at the output layer.

Tables 2–4 present the TPR, FPR and precision of different approaches for different types of attacks. It is seen in Tables 3 and 4 that Ensemble approach achieved less FPR and higher precision compared non-ensemble approach. DBN (equal hidden units in all layers) performs better than other individual DBN for normal traffic. In [24,47] Wei gao applied a decision tree based approach and achieved 93.1% detection accuracies mentioned in Table 1. The FPR from [24,47] has been reported in Table 3.

Ensemble of DBNs achieves lower FPR for MPCI and CMRI. In other cases, FPR is same with individual DBN. Proposed ensemble of DBNs also achieves better precision than other approaches for normal traffic. Performance for reconnaissance attack is same for all approaches.

Overall performance of the proposed ensemble of DBN is better than individual DBNs and existing approaches [24,47] using decision tree and support vector machines.

5. Conclusions

Modern industries are tremendously using IoT platform and connecting numerous smart sensors, IEDs (narrow band IoT compatibility) with its SCADA network for centralized control and better management in conjunction with cloud computing. This not only improves the production optimization to overcome the challenges of dynamic market situation, but also helps to monitor and respond toa safety situation in an improved way. The filed devices of industrial control systems (ICS), PLCs, IEDs and related protocols were initially designed to operate in isolated networks in which security and threat related issues were ignored. However, their integration with IoT platform and the Internet exposes the ICSs to significant security threats. Applications, devices and related

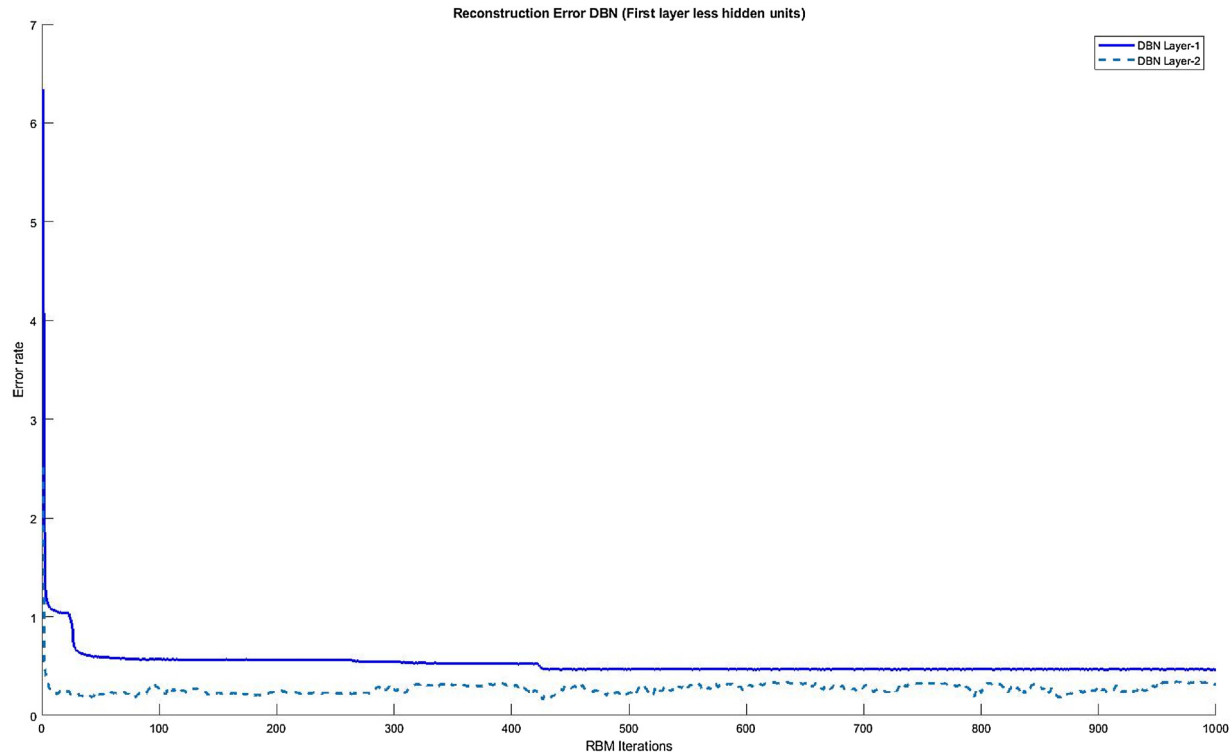

Fig. 6. Reconstruction Error in DBN (First layer less hidden units), RBM layer-1 and 2.

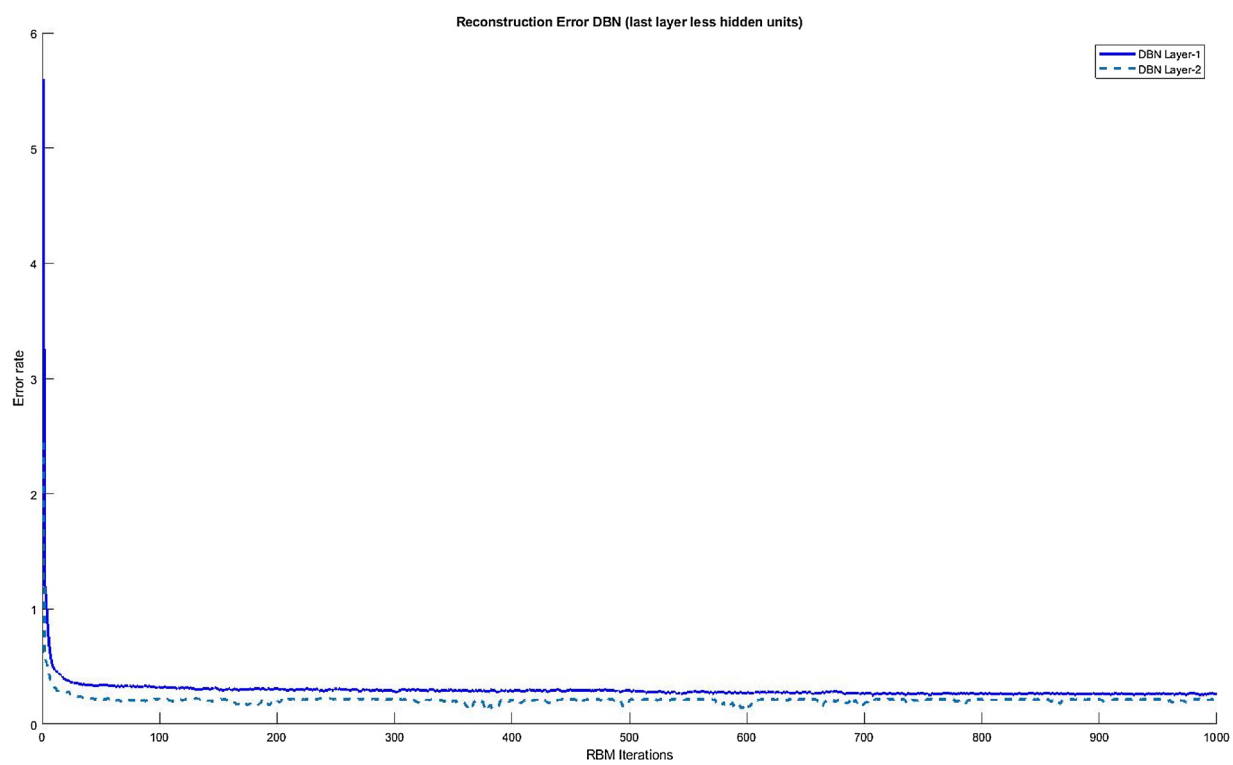

Fig. 7. Reconstruction Error in DBN (last layer less hidden units), RBM layer-1 and 2.

protocols of ICSs in modern industries have very specific functionalities and interaction within the networks which are different from the corporate computing systems. Therefore, conventional signature and API behaviour based anti-malware security tools are not sufficient to safeguard IoT-SCADA network of ICS.

In this research, we propose a secure SCADA architecture to protect ICS network from malicious attacks. We have proposed attack detection framework based on SCADA network traffic which are analysed and detected through ensemble based detection engine. Two different types of detection models are proposed based on deep belief network (DBN) and support vector machines (SVMs).

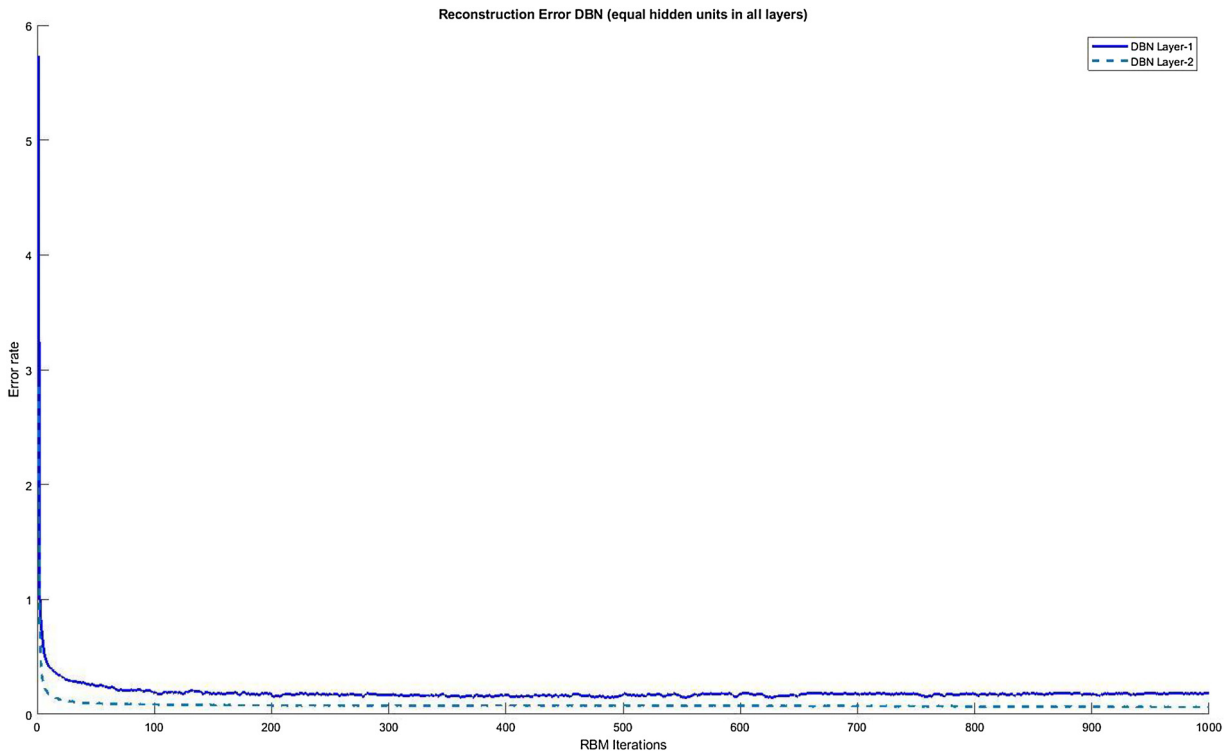

Fig. 8. Reconstruction Error in DBN (equal hidden units in all layers), RBM layer-1 and 2.

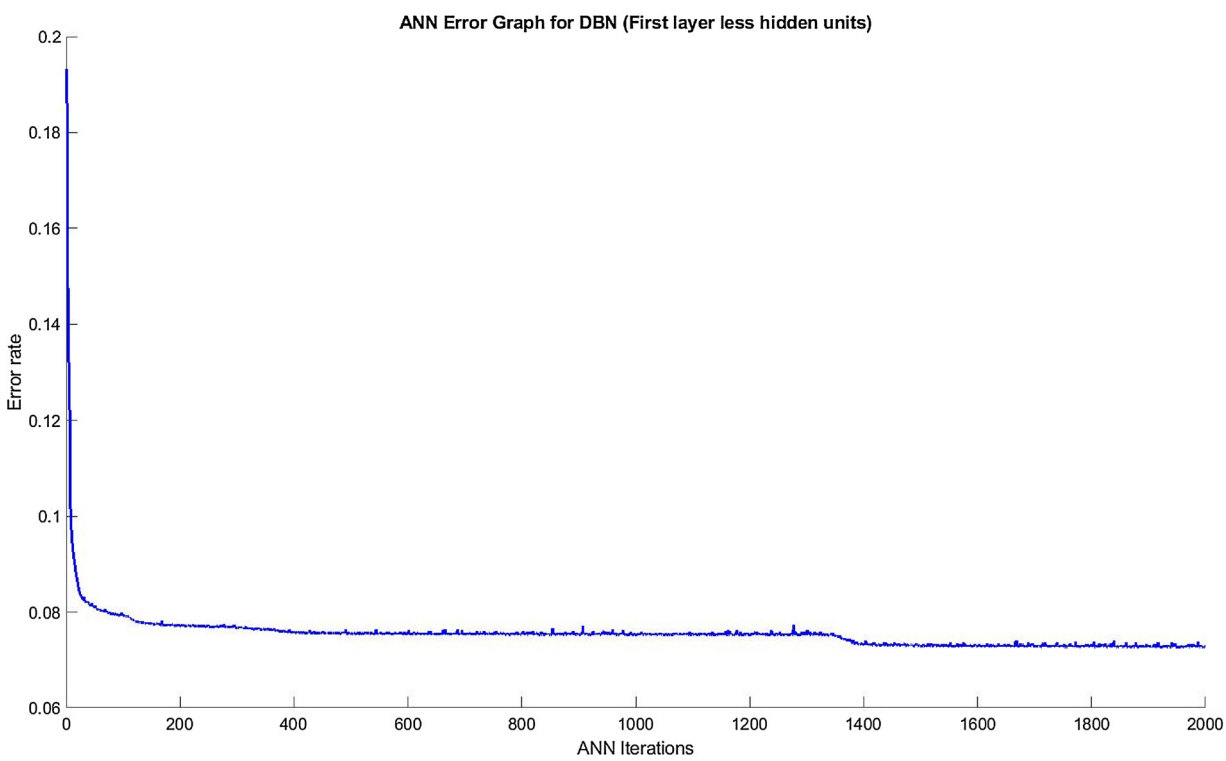

Fig. 9. ANN Error Graph for DBN (First layer less hidden units).

DBNs are popular discriminative approach for attack detection and can learn the hidden statistical structure of maliciousness in the network traffic through the composition of restricted boltzman machines (RBMs). However, for a large data set with redundant training samples, selecting an appropriate structure for DBN is a crucial problem. We propose ensemble approaches to DBN for SCADA network. Different structures of DBN are selected for building a set of attack detectors and then they are combined to construct an ensemble of DBNs for final detection. In addition to this, we also proposed an ensemble of classifiers based on a SVM [36] and bagging [35] algorithms. The proposed detection approach is verified using a real SCADA network traffic dataset. Experimental results

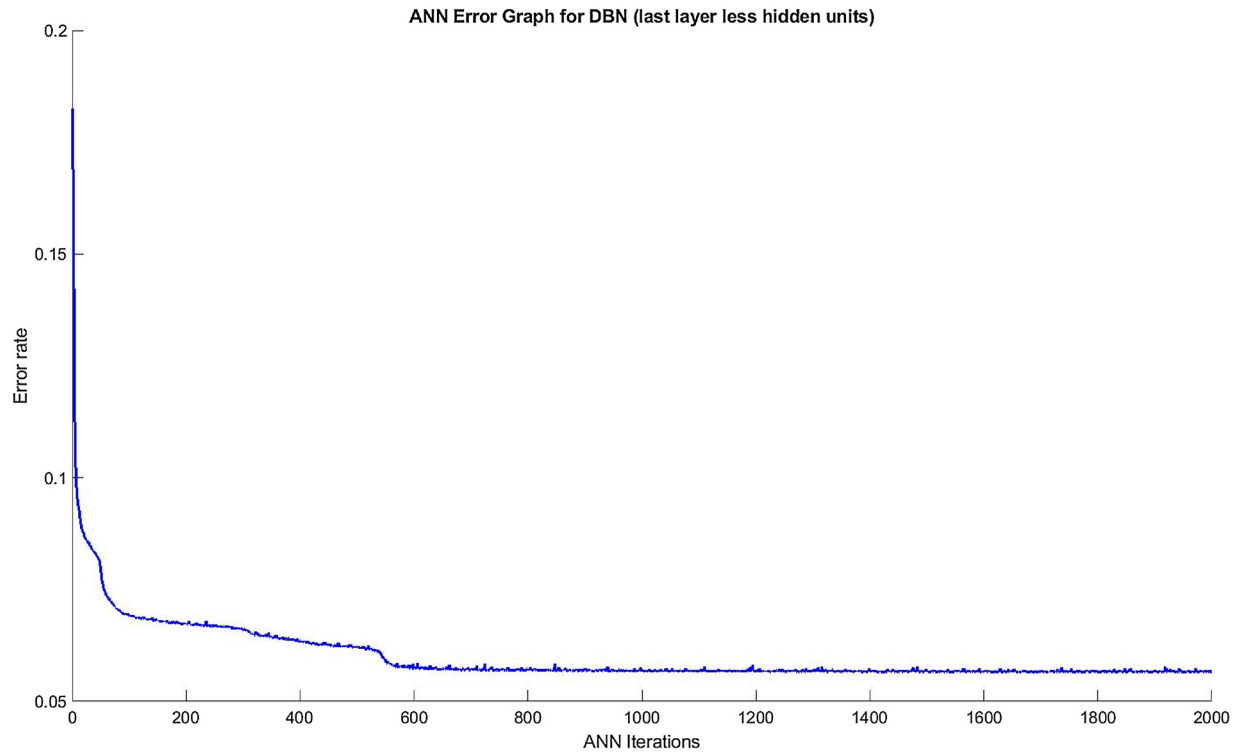

Fig. 10. ANN Error Graph for DBN (last layer less hidden units).

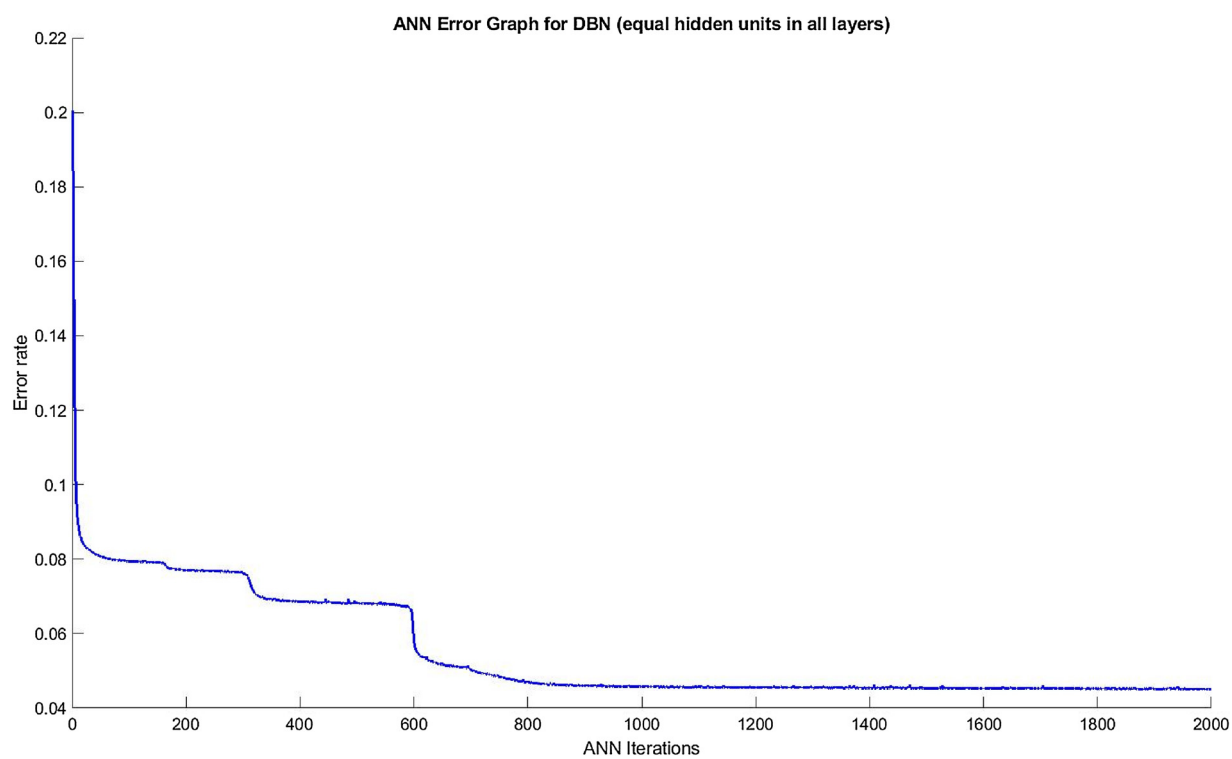

Fig. 11. ANN Error Graph for DBN (equal hidden units in all layers).

show that proposed ensemble approaches to DBN can overcome the structure selection problem of DBN and provide a high performance attack detection for secure operation of SCADA platform.

One of the limitations of our proposed approach is that it considers a neural network (NN) based classifier in the DBN framework. Although, DBNs are most popular techniques to extract features from raw data, its classifier is based on NN which may impact very high performance classification for the corresponding detection model. High performance classification is one of important requirements for SCADA, particularly for critical infrastructures. Features extracted from DBN can be used with other classifiers such as SVM for very high performance classification. Our experiments suggest

Table 2
True positive rate (TPR) for different detection approaches for different attack types.

Detection approaches (TPR) /Attack types	SVM	Ensemble of SVMs	DBN (equal hidden units in all layers)	DBN (last layer less hidden units)	DBN (First layer less hidden units)	Ensemble of DBNs
Normal	0.963	0.973	0.972	0.971	0.971	0.979
NMRI	0.000	0.000	0.000	0.000	0.000	0.000
CMRI	0.999	0.984	0.999	0.999	0.999	0.999
MSCI	0.949	0.949	0.949	0.949	0.949	0.949
MPCI	0.980	0.980	0.980	0.980	0.980	0.980
MFCI	0.958	0.958	0.958	0.958	0.948	0.958
DOS	0.979	0.979	0.979	0.979	0.979	0.978
Reconnaissance	1.000	1.000	1.000	1.000	1.000	1.000

Table 3
False positive rate (FPR) for different detection approaches for different attack types.

Detection approaches (FPR) /Attack types	SVM	Ensemble of SVMs	DBN (equal hidden units in all layers)	DBN (last layer less hidden units)	DBN (First layer less hidden units)	Decision Tree based approach [24,47]	Ensemble of DBNs
Normal	0.084	0.090	0.084	0.084	0.084	0.013	0.084
NMRI	0.000	0.000	0.000	0.000	0.000	0.008	0.000
CMRI	0.019	0.016	0.016	0.017	0.017	0.004	0.013
MSCI	0.000	0.000	0.000	0.000	0.000	0.007	0.000
MPCI	0.003	0.003	0.003	0.003	0.003	0.004	0.002
MFCI	0.000	0.000	0.000	0.000	0.000	0.002	0.000
DOS	0.000	0.000	0.000	0.000	0.000	0.00	0.000
Reconnaissance	0.000	0.000	0.000	0.000	0.000	0.006	0.000

Table 4
Precision for different detection approaches for different attack types.

Detection approaches (Precision) /Attack types	SVM	Ensemble of SVMs	DBN (equal hidden units in all layers)	DBN (last layer less hidden units)	DBN (First layer less hidden units)	Ensemble of DBNs
Normal	0.916	0.930	0.934	0.934	0.934	0.952
NMRI	0.000	0.000	0.000	0.000	0.000	0.000
CMRI	0.936	0.937	0.936	0.935	0.935	0.936
MSCI	0.974	0.974	0.974	0.974	0.974	0.974
MPCI	0.975	0.975	0.975	0.975	0.975	0.975
MFCI	1.000	1.000	1.000	0.984	0.968	1.000
DOS	1.000	1.000	1.000	1.000	1.000	0.992
Reconnaissance	1.000	1.000	1.000	1.000	1.000	1.000

that SVM can achieve high performance if more detailed features are provided. The other limitation of our work is that training of DBN in our proposed approach is not accomplished in real time settings. Future work may proceed to develop a parallel version of ensemble-DBN to find the optimal structures of DBNs for SCADA attack detection in ICSs. An FPGA based DBN model can be developed for proposed detection system for real time requirements. Also DBN can be used as a feature extractor and then can be combined with SVM. These can be accomplished in a future work.

Acknowledgement

The authors would like to extend their sincere appreciation to the Deanship of Scientific Research at King Saud University for its funding of this research through the research group project no. RGP-281.

References

[1] L.D. Xu, W. He, S. Li, Internet of things in industries: a survey, IEEE Trans. Ind. Inf. 10 (2014) 2233–2243.
[2] M. Basseville, Statistical method of change detection," robotics and automation, in: Unbehauen (Ed.), XVI, 2002, pp. 130–145.
[3] K.R. Hemangi Laxman, A.K. Gawanda, Bhattacharjee, Online monitoring of a cyber physical system against control aware cyber attacks, Procedia Computer Science, 4th International Conference on Ecofriendly Computing and Communication Systems 70 (2015) 238–244.
[4] K. Bu, M. Xu, X. Liu, Jiaqing Luo, S. Zhang, M. Weng, Deterministic detection of cloning attacks for anonymous rfid systems, IEEE Trans. Ind. Inf. 11/6 (2015) 1255–1266.
[5] M. Pajic, R. Mangharam, O. Sokolsky, D. Arney, J. Goldman, I. Lee, Model-driven safety analysis of closed-loop medical systems, IEEE Trans. Ind. Inf. 10/1 (2014) 3–16.
[7] A.A. Cardenas, S. Aminy, B. Sinopoliz, A. Giani, A. Perrigz, S. Sas, Challenges for securing cyber physical systems, in: Workshop on Future Directions in Cyber-Physical Systems Security, DHS, 2009, pp. 1–4.
[8] PandaLabs, in: Pandalabs Annual Report, panda Security, 2014 (Accessed March 18 2017 www.pandasecurity.com.
[9] Symantec, in: Internet Security Threat Report 2014, Symantec Corporation, 2014 (Accessed March 18 2017) www.symantec.com.
[10] E.K. Wang, Y. Ye, X. Xu, S.M. Yiu, L.C.K. Hu, K.P. Chow, Security issues and challenges for cyber physical system, in: Proc IEEE/ACM International Conference on Cyber, Physical and Social Computing, 2010, pp. 733–738.
[11] R. Mitchell, I.-R. Chen, Behavior rule based intrusion detection for supporting secure medical cyber physical systems, IEEE Trans. Dependable Secure Comput. 12/1 (2015) 16–30.
[12] National defense, in: Cyber Security for Advanced Manufacturing, a White Paper Prepared by, Technical Report, National Defense Industrial Associations, Manufacturing Division and Cyber Division, 2014.
[13] S. Naval, V. Laxmi, M. Rajarajan, M. Gaur, M. Conti, Employing program semantics for malware detection, IEEE Trans. Inf. Forensics Secur. 10/12 (2015) 2591–2604.
[14] S. Nourash rafeddin, E. Milios, D.V. Arnold, An ensemble approach for text document clustering using wikipedia concepts, in: Proceedings of the ACM Symposium on Document Engineering(DocEng '14), ACM New York, NY, USA, 2014, pp. 107–116.
[15] Shamsul Huda, Jemal Abawajy, Mamoun Alazab, Mali Abdollalihian, M.D. Rafiqul Islam, John Yearwood, Hybrids of support vector machine wrapper and filter based framework for malware detection, Future Gener. Comput. Syst. 55/C (2016) 376–390.

[16] W. Dong, X. Liu, Robust and secure time-synchronization against sybil attacks for sensor networks, IEEE Trans. Ind. Inf. 11/6 (2015) 1482–1491.
[17] D. Evans, The Internet of Things: How the Next Evolution of the Internet Is Changing Everything., Technical Report, CISCO White Paper, 2011.
[18] S. Huda, J. Abawajy, M. Abdollahian, R. Islam, J. Yearwood, A fast malware feature selection approach using a hybrid of multi-linear and stepwise binary logistic regression, Concurrency Comput. Pract. Exp. 29/ 23 (2017) 1–18.
[19] M. Alazab, S. Huda, J. Abawajy, R. Islam, J. Yearwood, S. Venkatraman, A hybrid wrapper-filter approach for malware detection, J. Netw. 9/11 (2014) 2878–2891.
[20] Yann LeCun, Yoshua Bengio, Geoffrey Hinton, Deep learning, Nature 521 (2015) 436–444.
[21] Geoffrey Hinton, Where do feature comes from? Cognit. Sci. 38 (2014) 1078–1101.
[22] Nitish Srivastava, Ruslan Salakhutdinov, Geoffrey Hinton, Modeling documents with a deep boltzmann machine, in: Proceedings of the Twenty-Ninth Conference on Uncertainty in Artificial Intelligence, 2013, pp. 616–624.
[23] J.J. Hopefield, Neural networks and physical systems with emergent collective computational abilities, Proc. Natl. Acad. Sci. 79 (1982) 2554–2558.
[24] Thomas Morris, Wei Gao, Industrial control system traffic data sets for intrusion detection research, in: Proceedings of International Conference on Critical Infrastructure Protection, 2014, pp. 65–78.
[25] C. Johnson, Securing the participation of safety-critical SCADA systems in the industrial internet of things, in: 11th International Conference on System Safety and Cyber Security (SSCS, London, UK, 2016, pp. 11–13, Oct 2016.
[26] Lorena Cazorla, Cristina Alcaraz, Javier Lopez, A three-stage analysis of IDS for critical infrastructures, Comput. Secur. 55 (2015) 235–250.
[27] P. Hehenbergera, B. Vogel-Heuserb, D. Bradleyc, B. Eynardd, T. Tomiyamae, S. Achichef, Design, modelling, simulation and integration of cyber physical systems: methods and applications, Comput. Ind. 82 (2016) 273–289.
[28] Zach DeSmita, Ahmad E. Elhabashya,b, Lee J. Wellsc, Jaime A. Camelioaa, An approach to cyber-physical vulnerability assessment for intelligent manufacturing systems, J. Manuf. Syst. 43 (2017) 339–351.
[29] Reza A. rghandeh, Alexandar von Meier, Laura Mehrmanesh, Lamine Mili, On the definition ofcyber-physical resilience in power systems, Renew. Sustain. Energy Rev. 58 (2016) 1060–1069.
[30] C.W. Johnson, M. Saleem, M. Evangelopoulou, R. Harkness, T. Barker, Albuquerque, USA, 21-25th August, Proceedings of the 35th International System Safety Conference (2017).
[31] Cristina Alcaraz, Rodrigo Roman, Pablo Najera, Javier Lopez, Security of industrial sensor network-based remote substations in the context of the internet of things, Ad Hoc Netw. 11 (2013) 091–1104.
[32] Ángel Manuel Guerrero-Higueras, Noemí DeCastro-García, Vicente Matellán, Detection of cyber-attacks to indoor real time localization systems for autonomous robots, Rob. Auton. Syst. 99 (2018) 75–83.
[33] Abdulmohsen Almalawi, Xinghuo Yu, Zahir Tari, Adil Fahad, Ibrahim Khalil, An unsupervised anomaly-based detection approach for integrity attacks on SCADA systems, Comput. Secur. 46 (2014) 94–110.
[34] Igor Nai Fovino, Andrea Carcano, Marcelo Masera, Alberto Trombetta, An experimental investigation of malware attacks on SCADA systems, Int. J. Crit. Infrastruct. Prot. 2 (2009) 139–145.
[35] L. Breiman, Bagging predictors, Mach. Learn. 24/2 (1996) 123–140.
[36] C. Cortes, V. Vapnik, Support-vector networks, Mach. Learn. 20/3 (1995) (1995) 273–280.
[37] Nicholas R. Rodofile, Kenneth Radke, Ernest Foo, Framework for SCADA cyber-attack dataset creation, Australas. Comput. Sci. Week (2017).
[38] B.R. Mehta, Y. Jaganmohan Reddy, SCADA SYSTEMS, chapter-7, in: Industrial Process Automation Systems, 2015, pp. 237–300.
[39] iTrust, Secure Water Treatment Testbed, 2015 (Accessed March 18 2017 https://itrust.sutd.edu.sg/research/testbeds/secure-water-treatment-swat/.
[40] S. Adepu, A. Mathur, An investigation into the response of a Water treatment system to cyber attacks, The 17th IEEE International Symposium on High Assurance Systems Engineering (HASE) (2017).
[41] S. Adepu, J. Prakash, A. Mathur, WaterJam: an experimental case study of jamming attacks on a Water treatment system, IEEE International Conference on Software Quality, Reliability & Security (2017).
[42] K.N. Junejo, D. Yau, Data driven physical modelling for intrusion detection in cyber physical systems, in: Singapore Cyber Security R&D Conference, 2016.
[43] B.H. Kim, S. Jo, Deep physiological affect network for the recognition of human emotions, IEEE Trans. Affect. Comput. (2018), http://dx.doi.org/10.1109/TAFFC.2018.2790939 (Early access) January.
[44] D. Wang, Y. Shang, Modeling physiological data with deep belief networks, Int. J. Inf. Educ. Technol. 3/5 (2013) 505–511.
[45] Xiang Zhang, Lina Yao, Quan Sheng, Salil Kanhere, Tao Gu, Dalin Zhang, Converting your thoughts to texts: enabling brain typing via deep feature learning of EEG signals, IEEE International Conference on Pervasive Computing and Communications (2018) 2018.
[46] Rijo Jackson Tom, Suresh Sankaranarayanan, IoT based SCADA integrated with fog for power distribution automation, Proceeding of 12th Iberian Conference on Information Systems and Technologies (CISTI) (2017) 21–24, June.
[47] Wei Gao, Cyberthreats, Attacks and Intrusion Detection in Supervisory Control and Data Acquisition Networks, PhD Thesis, in the Department of Electrical and Computer Engineering, Mississippi State, Mississippi, December, 2013.